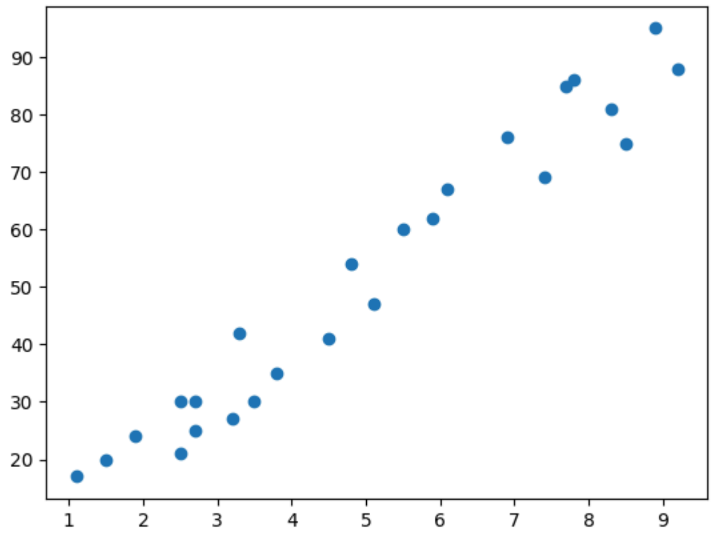
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
dataset = pd.read_csv('Student_scores.csv')
plt.scatter(dataset['Hours'], dataset['Scores'])
plt.show()
```




```
X = dataset.iloc[:,:-1].values
Y = dataset.iloc[:,1].values
```

```
print(Y)
```

```
[21 47 27 75 30 20 88 60 81 25 85 62 41 42 17 95 30 24 67 69 30 54 35 76
 86]
```

```
class Model():
  def __init__(self,learning_rate, iterations):
    self.learning_rate = learning_rate
    self.iterations = iterations

  def predict(self, X):
    return X.dot(self.slope) + self.const

  def fit(self, X, Y):
    self.m , self.n = X.shape
    self.slope = np.zeros(self.n)
    self.const = 0
    self.X = X
    self.Y = Y

    for i in range(self.iterations):
      self.update_weights()
    return self

  def update_weights(self):
    Y_pred = self.predict(self.X)
    dW = -(2 * (self.X.T).dot(self.Y - Y_pred)) / self.m
    db = - 2 * np.sum( self.Y - Y_pred ) / self.m

    self.slope = self.slope - self.learning_rate * dW
    self.const = self.const - self.learning_rate * db
    return self
```

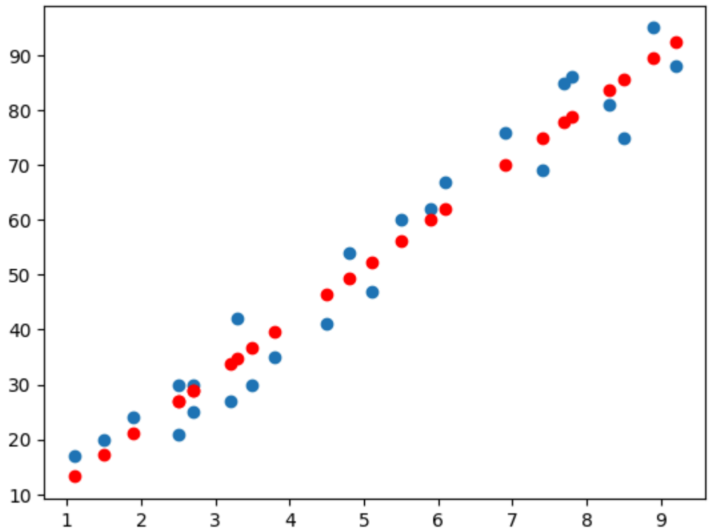
```
model = Model(learning_rate = 0.01, iterations = 1000)
model.fit(X,Y)
```

```
<__main__.Model at 0x7e80367ea4e0>
```

```
Y_pred = model.predict(X)
print(Y_pred)
```

```
[26.91171724 52.33687281 33.75695143 85.58515317 36.69062323 17.13281125
 92.43038736 56.24843521 83.62937197 28.86749844 77.76202838 60.1599976
 46.46952922 34.73484203 13.22124886 89.49671557 26.91171724 21.04437365
 62.1157788  74.82835658 28.86749844 49.40320102 39.62429503 69.93890359
 78.73991898]
```

```
plt.scatter(dataset['Hours'], dataset['Scores'])
plt.scatter(X, Y_pred, color = 'red')
plt.show()
print(model.slope)
print(model.const)
```




```
[9.77890599]
2.4644522714760995
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

datasets = pd.read_csv('50_startups.csv')
X = datasets.iloc[:,:-1].values
Y = datasets.iloc[:,1].values
```

```
print(X)
```

```
[[165349.2  136897.8  471784.1 ]
 [162597.7  151377.59 443898.53]
 [153441.51 101145.55 407934.54]
 [144372.41 118671.85 383199.62]
 [142107.34  91391.77 366168.42]
 [131876.9   99814.71 362861.36]
 [134615.46 147198.87 127716.82]
 [130298.13 145530.06 323876.68]
 [120542.52 148718.95 311613.29]
 [123334.88 108679.17 304981.62]
 [101913.08 110594.11 229160.95]
 [100671.96  91790.61 249744.55]
 [ 93863.75 127320.38 249839.44]
 [ 91992.39 135495.07 252664.93]
 [119943.24 156547.42 256512.92]
 [114523.61 122616.84 261776.23]
 [ 78013.11 121597.55 264346.06]
 [ 94657.16 145077.58 282574.31]
 [ 91749.16 114175.79 294919.57]
 [ 86419.7  153514.11      0.  ]
 [ 76253.86 113867.3  298664.47]
 [ 78389.47 153773.43 299737.29]
 [ 73994.56 122782.75 303319.26]
 [ 67532.53 105751.03 304768.73]
 [ 77044.01  99281.34 140574.81]
 [ 64664.71 139553.16 137962.62]
 [ 75328.87 144135.98 134050.07]
 [ 72107.6  127864.55 353183.81]
 [ 66051.52 182645.56 118148.2 ]
 [ 65605.48 153032.06 107138.38]
 [ 61994.48 115641.28  91131.24]
 [ 61136.38 152701.92  88218.23]
 [ 63408.86 129219.61  46085.25]
 [ 55493.95 103057.49 214634.81]
 [ 46426.07 157693.92 210797.67]
 [ 46014.02  85047.44 205517.64]
 [ 28663.76 127056.21 201126.82]
```

```
 [ 44069.95  51283.14 197029.42]
 [ 20229.59  65947.93 185265.1 ]
 [ 38558.51  82982.09 174999.3 ]
 [ 28754.33 118546.05 172795.67]
 [ 27892.92  84710.77 164470.71]
 [ 23640.93  96189.63 148001.11]
 [ 15505.73 127382.3   35534.17]
 [ 22177.74 154806.14  28334.72]
 [  1000.23 124153.04   1903.93]
 [  1315.46 115816.21 297114.46]
 [     0.   135426.92      0.  ]
 [   542.05  51743.15      0.  ]
 [     0.   116983.8   45173.06]]
```

```
from sklearn.linear_model import LinearRegression
regressor = LinearRegression()
```

```
from sklearn.model_selection import train_test_split
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size = 0.2, random_state = 0)
```

```
regressor.fit(X_train, Y_train)
```

```
▾ LinearRegression
LinearRegression()
```

```
y_pred = regressor.predict(X_test)
print(y_pred)
```

```
[182645.56  91790.61 110594.11  84710.77 101145.55 127864.55  65947.93
 152701.92 122782.75  91391.77]
```

```
for i, (pred,actual) in enumerate(zip(y_pred, Y_test)):
  print(f"Sample {i+1}: Predicted = {pred:.2f}, Actual = {actual:.2f}")
```

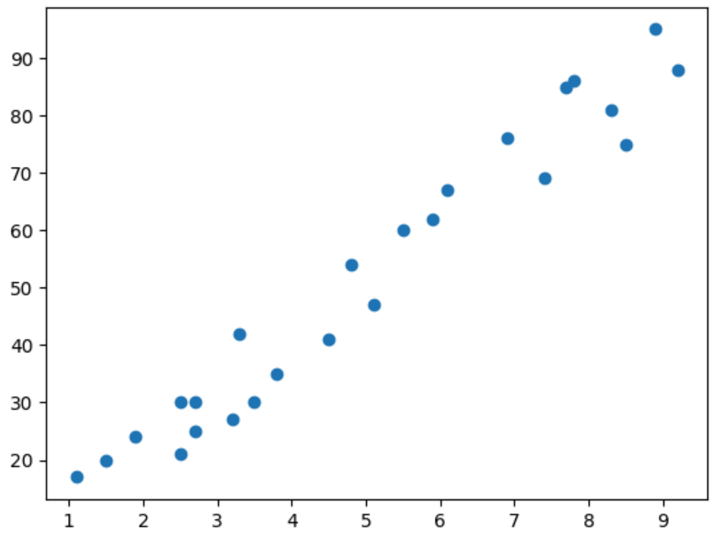
```
Sample 1: Predicted = 182645.56, Actual = 182645.56
Sample 2: Predicted = 91790.61, Actual = 91790.61
Sample 3: Predicted = 110594.11, Actual = 110594.11
Sample 4: Predicted = 84710.77, Actual = 84710.77
Sample 5: Predicted = 101145.55, Actual = 101145.55
Sample 6: Predicted = 127864.55, Actual = 127864.55
Sample 7: Predicted = 65947.93, Actual = 65947.93
Sample 8: Predicted = 152701.92, Actual = 152701.92
Sample 9: Predicted = 122782.75, Actual = 122782.75
Sample 10: Predicted = 91391.77, Actual = 91391.77
```

```
print("Coefficients: ", regressor.coef_)
print("Intercept:", regressor.intercept_)
```

```
Coefficients:  [ 2.36992324e-16  1.00000000e+00 -2.22044605e-16]
Intercept: -5.820766091346741e-11
```

Y = 2.36992324e-16*x1 + 1.00000000e+00*x2 + -2.22044605e-16*x3 + -5.820766091346741e-11

```
dataset = pd.read_csv('Student_scores.csv')
plt.scatter(dataset['Hours'], dataset['Scores'])
plt.show()
X = dataset.iloc[ :, : -1].values
Y = dataset.iloc[ :, -1].values
```




```
from sklearn.model_selection import train_test_split
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size = 0.2, random_state = 0)
```

```
from sklearn.preprocessing import PolynomialFeatures
poly = PolynomialFeatures(degree=3)
X_poly = poly.fit_transform(X)
print(X_poly)
```

```
[[  1.       2.5      6.25    15.625]
 [  1.       5.1     26.01   132.651]
 [  1.       3.2     10.24    32.768]
 [  1.       8.5     72.25   614.125]
 [  1.       3.5     12.25    42.875]
 [  1.       1.5      2.25     3.375]
 [  1.       9.2     84.64   778.688]
 [  1.       5.5     30.25   166.375]
 [  1.       8.3     68.89   571.787]
 [  1.       2.7      7.29    19.683]
 [  1.       7.7     59.29   456.533]
 [  1.       5.9     34.81   205.379]
 [  1.       4.5     20.25    91.125]
 [  1.       3.3     10.89    35.937]
 [  1.       1.1      1.21     1.331]
 [  1.       8.9     79.21   704.969]
 [  1.       2.5      6.25    15.625]
 [  1.       1.9      3.61     6.859]
 [  1.       6.1     37.21   226.981]
 [  1.       7.4     54.76   405.224]
 [  1.       2.7      7.29    19.683]
 [  1.       4.8     23.04   110.592]
 [  1.       3.8     14.44    54.872]
 [  1.       6.9     47.61   328.509]
 [  1.       7.8     60.84   474.552]]
```

```
model = LinearRegression()
model.fit(X_poly, Y)
```

```
▾ LinearRegression ⓘ ?
LinearRegression()
```

```
print(model.coef_)
print(model.intercept_)
```

```
[ 0.         -3.79705913  2.99874775 -0.19257785]
19.317706990502344
```

```
y_pred = model.predict(X_poly)
print(y_pred)
```

```
[25.55820366 52.4044897  31.56390366 85.43535572 34.50588455 19.71935048
 88.24070983 57.10586101 84.27233622 27.13600857 79.9579611  61.75002055
 45.40692604 32.52310457 18.5131056  87.29327399 25.55820366 21.60788253
 64.02753655 77.39372855 27.13600857 48.88540147 37.62366789 72.62482165
 80.75625387]
```

```
plt.scatter(X, Y, color='blue')
plt.scatter(X, y_pred, color ='red')
```

```
plt.title('Polynomial Regression')

plt.show()
```